

TGI: Topological Geometric Intelligence

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Abstract

Topological Geometric Intelligence (TGI) is a discrete, source-owned approach to forming and revising inspectable knowledge from raw character and event streams. It preserves the NINMENI invariant of one static identity per Unicode character while deriving TGI-specific spatial addresses, C/L-band ownership, phase admission, interventional relations, deterministic resolution and state-owned execution. We specify the objects and operators as one connected system and test them using shared-prefix reconstruction, source and phase counterexamples, signed intervention traces, complete-frontier decisions, actual feedback with cold recovery, and locked scale protocols. The resulting claims are attached to their own falsifiers: exact traversal does not by itself establish a world referent, while a source-valid relation requires observational or interventional support. This article consolidates the mechanism, experiment and falsifier chain in one auditable technical account.

Keywords: Topological Geometric Intelligence; NINMENI; NMU; discrete geometry; crystallization; source provenance; interventional learning; deterministic resolution.

1. Introduction

The research question is whether a machine can form, revise, and use knowledge through discrete identity and geometry while preserving an inspectable chain from raw experience to answer. The NINMENI NMU study fixes one character as one static identity (Leunufna, 2026a). MULTIPITA gives a principle of band-owned computation without replacing that identity (Leunufna, 2026b). The Emylton and Crystal Memory studies distinguish organizational state from persistent memory and retrieval (Leunufna, 2026c; 2026d). TGI adopts those principles, then derives its own geometric, source-valid decision operators. It does not import a VEYRA NYRA model.

For a raw stream of Unicode scalars, a static ID belongs to a character, while a source, event time and position belong to an occurrence. A discrete address makes an occurrence retrievable; a bond and phase receipt determine whether traversal is admitted; organization measures support across source-valid experience; intervention distinguishes a world relation from linguistic repetition. Resolution must examine the complete relevant frontier, retain counterevidence and report incomplete work distinctly from an absent path. These are successive proof obligations rather than interchangeable scores.

The implementation excludes subword identities, gradient descent, backpropagation, softmax scaled dot-product attention, and continuous vector embeddings as TGI learning or resolution operators. The use of arrays, indexes, hashes and compiled code is an implementation choice constrained by identity and provenance preservation. The paper gives the definitions, falsification protocols, experimental outcomes and execution costs of the integrated system.

2. Mechanisms and formal method

2.1. Character identity and discrete lattice

Identity and occurrence definitions

Let U be the declared admitted Unicode scalar space. The TGI codec is $l: U \rightarrow N$, $l(c) = \text{ord}(c) + 16$. The low 16 integers are not character IDs. For admitted scalars, the map is injective and static. A surrogate is rejected before it can become a stored identity. Input is neither NFC-normalized nor segmented. This is an explicit TGI instantiation choice. In the NMU study (Leunufna, 2026a), NINMENI's measured corpus and 10,240-slot registry use a declared normalization and hash-bound table; importing that registry's implementation into TGI would be an unjustified architectural shortcut. What TGI takes from the NMU study (Leunufna, 2026a) is the identity **axiom**, not its corpus-specific table.

For a source s , frame f , and position i , define the occurrence $o = (s, f, i, l(c_i))$. The projection $\text{identity}(o) = l(c_i)$ forgets event ownership, so two occurrences can share an identity without being interchangeable. The ordered stream is $O_f = (o_0, \dots, o_{T-1})$; its chronology and raw source must survive checkpointing. A valid bulk-native operator may process several occurrences together, but it must project back to every original identity and position in the same order. This is a

stronger invariant than equality of the final rendered string.

The distinction matters for repeated letters and shared prefixes. In the sentence *SISTEM TGI* ..., both *S* occurrences have the same static ID but not the same occurrence address. The *S* at the start of the other sentence also has that ID, while its source lane differs. Neither a numerical ID nor a single coordinate carries a meaning label such as *safe* or *reactor*.

Discrete layout and route certificate

The versioned foundation layout places occurrence i of the n th distinct frame at

$$P(n,i)=(2+i, 1, 0, 3+5n, (5-3n) \bmod 11).$$

The first coordinate orders characters; the fourth and fifth create source lanes. This is a constructive address allocation, not a proof that Euclidean or hyperbolic geometry by itself forms knowledge. For an admitted frame, a forward adjacency bond joins $P(n,i)$ to $P(n,i+1)$ only within its own lane. The route certificate records source, ordered occurrence identities, addresses, bond validity, and an explicit terminal marker. A reader returns the original character stream only after reaching the terminal marker through checked bonds. Three outcome categories are kept distinct:

Outcome	Necessary observation	Interpretation
COMPLETE	every ordered occurrence and terminal bond verified	exact reconstruction of a stored route
NO_PATH	requested anchor is absent from valid state	no certified continuation
INCOMPLETE	a route is present but broken or stopped by a ceiling	no complete output claim

The codec can decode each visited ID back to its scalar. A complete certificate therefore implies exact output character by character for that stored route. It does not imply factual truth of the decoded sentence. The no-path rule is a decision about the current geometric memory, not a universal statement that a sentence is impossible.

Formal invariants and a constructive argument

The character mapping is injective because adding a fixed offset to two different Unicode scalar numbers cannot make them equal. This is an identity property, not a statistical observation. The occurrence projection is not injective: two events may share $l(c)$. Thus no proof about event ownership may be based on ID equality alone. The address allocation must additionally bind the frame number, within-frame position, and source journal. For two frames $n \neq m$, their fourth coordinates differ by $5(n-m)$; hence their lanes cannot coincide under the declared layout. For two positions in the same frame, first coordinates differ. This establishes occurrence separation inside the admitted finite frame set independently of the wording of the sentences. The fifth coordinate is an additional consistency check, not the only source of uniqueness.

For exact reconstruction, take the route whose bonds connect the ordered occurrences $o_0, \dots, o_{(T-1)}, END$. The base case reads $l(c_0)$ from o_0 . At step $i < T-1$, the validated forward bond has the same frame lane and the unique successor $P(n,i+1)$; decoding it appends $c_{(i+1)}$. Induction gives the original scalar sequence after T occurrences, provided the terminal bond is present. If a bond is missing, the induction premise fails and the route cannot emit a complete result. This is the scope of the constructive theorem. It does not establish that a different unstored sentence should have the same result.

The absence decision is equally precise. An unformed anchor has no initial valid occurrence and therefore no initial route certificate. Returning an empty result is a property of the current state; returning one of the two stored sentences would be a counterexample to the contract. A safety ceiling interrupting a nonempty route is recorded as *INCOMPLETE*, avoiding the mistake of conflating unobserved data with a proof of absence.

2.2. Band-owned phase and organization

Band ownership without identity pooling

For a frame $f=(c_0, \dots, c_{(T-1)})$, fast bank $j \bmod 8$ owns the adjacency witness between positions j and $j+1$. Ownership answers *where the check is maintained*, not which characters exist. Every character still has its own static NMU ID and ordered event. The eight slow banks receive temporal coverage through binary carry. A completed block can be incorporated into a larger block while the raw occurrence order and source remain recoverable. An empty bank is not filled with inferred support.

The invariant is stronger than equal output after processing. Given a stored receipt, an independent checker must recompute the expected owner of each adjacency and temporal block from the original frame length and event order. A receipt that attributes a valid-looking value to the wrong bank is invalid. This protects against the common failure in which

a plausible scalar is accepted while the mechanism purported to produce it never ran. The C/L organization here is a TGI structural implementation; it is not the trained channel-partitioned MP2 cell of (Leunufna, 2026b).

Two levels of density and the lock

The foundation occurrence law uses c_i, l_i in $\{0, 1\}$ as validity witnesses for the local and temporal paths:

$$\Omega_i = (c_i + l_i)/2, \text{ with } \Omega_{crit} = 1.$$

An occurrence becomes structurally eligible only if both checks pass. The versioned xi_i lock additionally requires a complete frame, an owned coordinate, and a unique occurrence. This is a certificate bit or scalar projection of the checked state, not an independent source of truth. In particular, a high value caused by a normalization invariant is not a health metric and cannot license a world claim.

An organization H is inferred from multiple raw episodes. Stable intervals remain literal NMU occurrences. Intervals that vary together may become binding axes, but the axis is an organizational relation over occurrences, not a new token identity. An episode supports H only when its complete raw sequence has exactly one valid interpretation under H . Let $D_k(H)$ be the set of distinct realizations of axis k among such episodes. The frozen candidate measure is $\Omega_H = \min_k |D_k(H)|$ with threshold three. Ω_H and Ω_i have different domains and should never be substituted for one another in a derivation or in a plot.

The phase decision is therefore source-bound. A candidate with enough realizations cannot be promoted if a contributing source has a malformed frame, false C/L ownership, invalid receipt, or counterexample. The checker stores the original evidence scope used for the decision, making a later revocation and re-evaluation possible. No negative raw event is deleted to make a crystal appear stable.

Phase verification algorithm and failure analysis

The independent phase check can be described without relying on a particular Python data structure. It first decodes the declared raw frame and rejects invalid scalar identities. It then reconstructs every fast-bank owner from its adjacency index and every slow-bank block from chronological binary carry. It compares these reconstructed structures with the stored receipt. Only after this equality holds does it compute occurrence validity and the candidate lock. At the organization level it enumerates source episodes, requires complete unambiguous interpretation, derives the set of distinct axis realizations, and checks the current source-validity state. A later counterexample forces a new evaluation over the preserved event history.

This order matters causally. If the checker read an asserted xi first and used it to excuse a malformed C/L record, it would make the claimed mechanism self-certifying. If it counted every episode that resembles an axis but has two valid interpretations, Ω_H would be inflated by an undecided source. If it removed an opposing source, the subsequent crystal would appear stronger only because evidence was destroyed. The tested implementation rejects those routes and exposes why promotion is withheld.

Evidence object	Unit of observation	What it may justify	What it cannot justify alone
c_i, l_i	one occurrence check	local and temporal ownership	factual truth of text
Ω_i, xi_i	source/frame phase receipt	structurally readable occurrence	organization across episodes
Ω_H	distinct unambiguous axis realizations	candidate organization	correct world referent
counterexample	later original source event	revocation or context refinement	deletion of older source history

The table also prevents a nomenclature failure. NINMENI's Emylton state uses Ω for organization history and xi as a projected observation; TGI's occurrence and organization receipts use similar letters but own different objects. The shared principle is that organization and projection must be separated. Equality of symbols is not a derivation from Emylton.

2.3. Interventional spatial relations

Raw edit contrast and transport

For a port p , an event is $e_t(p)=(source_t, l(a_t), l(before_t(p)), l(after_t(p)))$, where l maps each scalar to its fixed NMU ID. The contrast operator first identifies the valid one-NMU edit kind: *STABLE*, *SUBSTITUTE*, *INSERT*, or *DELETE*. Ambiguous insertion/deletion positions are all retained. Multi-atom changes cannot be silently simplified into a one-atom response. The earlier signed three-port audit is useful precisely because some inputs failed these conditions and returned *NO_PATH* rather than being selected as positive evidence.

The first response-class rule compared edit kind and absolute offset. A falsifier showed it separated $T0 \rightarrow T1$ from $LONG:0 \rightarrow LONG:1$ even though both changed 0 to 1, and could equate different changed atoms at the same offset. The V4 transport rule instead uses the changed identity:

$\tau(SUBSTITUTE)=(SUBSTITUTE, old_ID, new_ID);$

$\tau(INSERT)=(INSERT, inserted_ID);$

$\tau(DELETE)=(DELETE, deleted_ID);$

$\tau(STABLE)=(STABLE).$

The complete before/after frames, positions, and source remain in the certificate. Discarding the absolute offset from the **class key** is not discarding it from evidence. The operator makes a limited invariance claim: stable textual wrappers may change while the local NMU edit is conserved. It does not equate arbitrary paraphrases or unseen sensor transformations.

Class formation and phase law

Let $h_t(p)$ be the chronological sequence of certified τ values for port p . Two ports are in the same observational class at time t if their histories agree under the declared transport rule. This equivalence is operational: it says available interventions have not separated them. Members retain distinct addresses and raw histories.

For a candidate class and action, C support counts distinct raw contexts before a class-wide change. L requires (i) a same-action stable control and (ii) an appropriate comparator from another action while the candidate class remains stable. The candidate reaches an Omega support threshold only with distinct contexts. xi locks only one eligible action-class pair; ties or missing controls return *NO_PATH*. These requirements were frozen before the signed positive capture. They prevent repeated identical observations or a target label from masquerading as independent support.

The phase transition is reversible. Suppose class $\{p,q\}$ is *CRYSTAL* and the next actual action changes q but not p . The event must return *REVOKED*. A V1 falsifier showed that immediate promotion of $\{q\}$ from old support was wrong. The repaired law starts a new evidence epoch at the fracture. The old events remain in the certificate, but only fresh support, controls, and comparator evidence may crystallize $\{q\}$. This is a specific mechanism of source-preserving forgetting of a decision right; it is not deletion of experience.

Active choice without an effect oracle

After an ambiguous class crystallizes, TGI may seek a separating experience. Given a verified class certificate and a finite available-action catalogue, the V5 selector computes which action strings have already occurred in the current epoch. It preserves all untried alternatives and selects one by lexicographic NMU-ID order. The catalogue is an interface affordance, not a complete ontology of world actions. The certificate explicitly reports *effect_predicted=false* and *separation_claimed=false*. If no action remains, the outcome is *NO_UNTRIED_ACTION*, not a proof that two physical entities are identical. Both source certificate and choice are independently checked.

Event-level decision trace and causal reading

Event range	Raw action	Class evidence state	Native decision
1-4	a,a,a,a	distinct changing contexts and a same-action control accrue	<i>NO_PATH</i> at each prefix
5	b	cross-action comparator closes eligibility	two-port <i>CRYSTAL</i>
6	c	port responses diverge for the first time	<i>REVOKED</i>
7-10	c,c,c,c	support and a control accrue after the fracture epoch	<i>NO_PATH</i> at each prefix
11	b	comparator for the fresh candidate is observed	one-port <i>CRYSTAL</i>

The significance of event 6 is not that the letter *c* possesses a predefined semantic role. It is that a measured action changes one port's response history while leaving its former class mate different. If the collector had supplied *target=q* to native input, the same status sequence would not prove endogenous class formation. The protocol instead stores the port/component mapping as evaluator-only metadata and runs a label-invariance control. Likewise, the new one-port crystal at event 11 is not inherited from the pre-fracture support: the event trace checks the epoch boundary and fresh support explicitly.

An observational class is therefore a **quotient of recorded response histories**, not a metaphysical equivalence relation over hidden physical objects. It is symmetric and transitive for the fixed history because equality of verified *tau* sequences is symmetric and transitive. It can refine after a new event; that change is not a contradiction in the definition because the time index has changed. Treating a past class as immutable would be the actual contradiction. The phase lock gives the class a current reading right and the fracture law removes that right when its premises cease to hold.

2.4. Deterministic resolution and continual revision

Source-derived operators and context guards

An actual transition records a complete before frame, raw action, after frame, source ID, port set, and ordering. Minimal context families are computed from original measured relations. A guard is a conjunction over the ports in one family. A transition law may be an exact literal case for one observed before frame or a whole-before operator such as prepend, append, interleave, or repetition. The operator is not inferred from the surface similarity of two examples alone. Its admission requires:

1. a phase-valid single-axis whole-before organization; 2. every axis bound and every original organization agreeing with the proposed operator; 3. exact action applicability and replay of every original literal source; 4. agreement of all applicable minimal context families on the **full concrete output**, not merely on an abstract region name.

If an operator, guard, or literal case is missing, the decision records a specific acquisition obligation. It does not create an implicit wildcard. The V116 guarded-word decision details the admission conditions. This differs from memorizing a list of source-output pairs: the admitted operator can act on a new whole before string under a proved guard, while an unproved operator remains unavailable even if it would happen to predict a test answer correctly.

Finite proof dictionary and complete frontier

Let *D* contain all nonempty substrings of original source guards and declared goals. The V116 proof treats an input outside *D* as nonmember under an operator that preserves that whole input; a constant operator can instead return to a known exact word. This gives an exact finite congruence for the source-proved guarded-word system, not an unrestricted theorem about all future language. The state for planning is the complete joint collection of port regions and concrete values. All available actions, reachable regions, self-loops, return edges, and shortest alternatives are enumerated and checked independently (Leunufna, 2026a; 2026d).

A frontier certificate owns the search result. It may identify a complete supported goal path, a shortest acquisition request, multiple unresolved alternatives, or a closed no-path result within the declared state/action system. A ceiling that stops enumeration yields incomplete status, not a synthetic theorem of impossibility. Context-changing actions update the entire joint state before the next step is selected. This is important when two ports are correlated in the current experience but an actual separating action later shows they require distinct context families.

For an answer *y*, the constructive claim is conditional: *Certified(q, S, y)* requires that every selected edge have a source-valid operator or literal witness, all competing applicable families agree on *y*, the complete route is within the verified frontier, and the current source history contains no revoking counterexample. If any premise fails, the system must preserve the obligation or refuse the answer. This certificate is the content of deterministic resolution; merely choosing the largest bond score would not satisfy it.

Actual experience and knowledge renewal

Selection initiates a real action through the external interface. The returned before/after frames are admitted as original NMU events, not as an expected-output fixture. Unexpected effects are retained as knowledge or opposition; they can interrupt the current plan and trigger a new frontier. Repeated support does not delete the counterexample. Later contexts may separate apparently conflicting laws, but that separation itself needs an actual distinguishing witness.

The owned continuation cycle performs verified prefix, actual continuation, learning, complete-history renewal, and atomic commit. The checkpoint contains the original event history and the proof needed for a fresh-process recovery. A malformed input, stale external digest, or certificate that disagrees with the raw events is rejected before partial mutation

becomes visible. This avoids a common form of false continual learning: a process whose in-memory state seems updated but cannot reproduce the same decision after restart.

The V119 local acquisition integration adds source-owned updates above the prior native runtime and keeps evaluator-label substitutions outside the native input. The older integration decision is not used to assert every later capability by inheritance. The current source archive and its extracted phase are the reproducibility anchor for the latest packaged runtime; older evidence retains its own frozen binding.

Resolution procedure and certificate decomposition

At a fixed current source state, the procedure first verifies the complete original event ledger and phase validity. It then derives every minimal context family and literal law from those events. Candidate whole-before operators are admitted only after the four source tests in Section 2. The proof dictionary is built from the finite guards and declared goals, and all reachable joint-state edges are enumerated for the declared available actions. The frontier checker recomputes the same graph from source laws, compares all shortest alternatives, and checks each proposed goal path's original NMU lineage. The selector may then choose one certified action or an acquisition obligation. After an actual executor result returns, the complete process repeats on the enlarged ledger.

This order avoids a subtle circular proof. The expected answer must not be fed into the organization that later “proves” the answer. Nor may a selected action's anticipated effect be substituted for the actual before/after frames. The source-derived law and measured effect have distinct provenance fields. When they disagree, the disagreement is counterevidence and changes the knowledge decision. A fresh process recovery verifies the same sequence without consulting an in-memory shortcut or a hidden evaluator label.

Certificate layer	Owned premise	Typical falsifier
source/phase	raw event and valid organization	damaged or stale source receipt
operator	all bound axes and exact literal replay	forged universal-input witness
context	every applicable minimal family agrees	newly separating actual action
frontier	all reachable declared edges and shortest alternatives	omitted self-loop or return edge
actuation	actual action/result chronology	expected outcome substituted for observed outcome
checkpoint	atomic ledger and certificate	wrong external digest or partial write

The statuses have operational consequences. *CERTIFIED* permits the declared route's output or next action. *NO_PATH* means no supported route in the completed declared graph. *REVOKED* means an earlier route's premise has been contradicted or its source has lost validity. *INCOMPLETE* means search, capture, or verification stopped before a decision could be proved. Combining these into a single “low confidence” number would lose information needed for the next experiment.

2.5. Owned spatial execution and scale

V10 diagnosis and V11 ownership correction

The V10 function profile attributed a dominant cumulative cost to *copy.deepcopy* of the full certificate on repeated *observe* calls. At 256 sources, the recorded cumulative deepcopy time was 32.245 s for the V9 control and 32.679 s for the V10 cycle under *cProfile*. Those figures are attribution diagnostics, **not** untraced throughput comparisons. The candidate V11 repair changes internal ownership access only. It does not remove the defensive public property, discard raw history, weaken the checker, or change an event's NMU identity.

An explicit parity probe compares V10 source distribution bytes with the V11 candidate. It records matching checkpoint hash, certificate hash, byte count, resolution, and cold recovery for the declared synthetic probe. The full regression and extracted package tests guard against effects outside the microbenchmark. These controls are required before reading the timing table as evidence of a valid optimization.

Frozen scale design

The V11 protocol was written before collecting three new untraced replications; a fourth traced run is kept separate. Every workload uses 256-NMU frames, a fixed count of sources in $\{32, 64, 128, 256\}$, and the same functional controls. There are two panels: the spatial V9-form decision and the temporal-phase decision. The comparator is a locked V10 baseline identified by SHA-256. For each size/panel the report retains individual append times, median append, full-replay times, checkpoint size, peak allocated bytes, PID, and hash. The criterion was lower median append than the baseline at all eight points, with replay within 20% of baseline.

The principle of *matched workload* matters. A *cProfile* run is not used as an untraced timing replicate; a source count of 256 is not presented as a whole production corpus; and a lower append time is not taken to mean the scientific relation learned is more accurate. The same distinction appears in the MULTIPITA paper (Leunufna, 2026b), which separates arithmetic, isolated kernels, full trainer, and training observation. TGI applies that discipline to its own operators.

3. Experimental protocols and results

3.1. Character identity and discrete lattice

Method and falsifiers

The frozen user acceptance case contains two raw streams:

1. *SISTEM TGI AMAN DARI HALUSINASI*; 2. *SISTEM REAKTOR NUKLIR MEMBUTUHKAN PENDINGIN*.

The two streams share the first six characters. The accepted anchors are the first *S* in lane A and the first *S* in lane B; a third unformed anchor is an absence control. Successful evaluation requires exact reconstruction of both strings, zero cross-lane jumps after *SISTEM*, and no output from the absent anchor. The acceptance report executes the route and the broader project regression in the same chain.

Output equality is not sufficient to verify spatial ownership. A historical checkpoint fault sorted sources before replay, producing the same text while changing which frame obtained which lane. The spatial retention decision therefore freezes the original raw insertion journal and verifies the lattice after fresh-process recovery. Frame equality is checked from the complete NMU sequence rather than trusting a potentially colliding digest. A broken bond, modified coordinate, or mismatched receipt must fail validation rather than fall back to a plausible text completion.

The acceptance test is deliberately sharp: shared prefixes make a shallow character lookup appear sufficient for the first few positions, but only a lane-bound route can preserve the correct continuation. Conversely, an absent anchor should not be mapped to the closest observed source. The reader has no softmax sampling step that could hide absence behind an output string.

Results and inference

The source acceptance report records both exact reconstructions and the absent-context decision. The checkpoint-order falsifier and fix are retained in . The phase-validity requirements required before a frame contributes are defined separately in ; they prevent an invalid source receipt from silently serving as a route. The source distribution report checks that the relevant implementation and tests can be reproduced from extracted package bytes. These evidence classes should not be collapsed: a regression checks behavior, a checkpoint check tests state transport, and a source archive check tests reproducibility.

This establishes the TGI substrate and exact route behavior for the declared states. It does not claim that $P(n,i)$ alone discovers a physical referent or that the original integral expression for x_i in the original TGI research statement has been derived from this layout. That expression is a source hypothesis. The tested placement law is an explicit implementation whose premises and consequences are visible.

Test matrix and state provenance

Probe	Perturbation	Required certificate consequence
same prefix, two sources	trigger first <i>S</i> under each lane	exact respective full stream, no cross-lane edge
third anchor	request a lane never formed	<i>NO_PATH</i> , no guessed phrase
source order swap on recovery	reorder journal while keeping text	reject changed coordinate ownership

Probe	Perturbation	Required certificate consequence
damaged bond or terminal	remove one required route edge	<i>INCOMPLETE</i> or invalid certificate, never partial text labeled complete
Unicode distinction	feed different scalar sequences with similar display	preserve distinct IDs and raw order

The journal and the certificate carry different roles. The journal is provenance of what entered and when; the certificate is a derivation from that journal under current rules. A certificate recovered without the underlying raw history would be insufficient to test whether source order changed. Conversely, merely retaining the journal does not prove that the current lattice still obeys the placement law. Both are checked after recovery. This mirrors the NINMENI papers' distinction between a declared identity rule and a measured state, while keeping TGI's own discrete addressing rule explicit.

3.2. Band-owned phase and organization

Falsification protocol

The tests challenge four separable properties:

Challenge	Required outcome	Why it matters
C/L owner altered, output unchanged	reject receipt	output parity alone misses wrong ownership
incomplete or fabricated bank support	no crystallization	prevents support from an empty path
counterexample after crystal	revoke affected reading right	tests knowledge validity through time
old certificate after new evidence	reject or revalidate against full history	prevents stale confidence

The raw-organization and phase acceptance reports carry the first versioned evidence. The source distribution rebuilds and evaluates the phase from extracted archive bytes, rather than relying on a workspace whose history can change. Historical negative audits are retained in the long-form TGI evidence ledger . The experiment classification follows (Leunufna, 2026b): a structural check, a regression pass, a systems measurement, and a functional capability claim are distinct kinds of evidence.

Results, interpretation, and successor interfaces

The accepted raw-organization report records nine gate decisions; the phase report records source-valid phase, counterexample, and polarity checks. The source archive passes its material/hash gates and runs both phase and distribution verifiers from extraction. These results establish a reproducible mechanism for source-valid structural promotion and withdrawal on the tested inputs. The acceptance chain also tests the exact route consequence when the source and frame are valid.

The earlier TGI formulation writes a general resonance accumulation for Omega and an integral expression for xi. The equations above are operational laws for this implementation. Matching names or dimensions would not prove mathematical equivalence to the exploratory manifesto expressions. The paper therefore states the implemented operator and its falsifier directly. This choice makes the chain testable: the relation analysis below may form relations only from events whose source and phase are certified. It may not relabel the phase scalar as a semantic predicate.

What the results test

The organization report's nine gates and the phase report's six checks are meaningful only with their input scope, exact source bytes, and negative controls. The source-distribution verifier adds a different guarantee: it checks that the same files appear inside the archive with the expected hashes, then executes phase and distribution checks from the extracted tree. It cannot by itself enlarge the functional input class beyond the declared contracts. The functional consequence relevant to this article is narrower and constructive: a route can be read only through source-valid crystals; when a counterexample revokes that source or organization, the old certificate cannot continue to authorize resolution. The actual query-level effect is checked by the separate resolution analysis.

3.3. Interventional spatial relations

Frozen physical protocol and controls

The strongest local witness is a newly created state file with initial $\{x:0,y:0,z:0\}$ and three opaque sensor ports. The collector's fixed action schedule is a,a,a,b,c,c,c,c,b . a increments x to ceiling three, b increments y to ceiling two, and c increments z to ceiling four. The sensor frames are $T(x)$, $VIEW:(x+z)$, and $D(y)$. Each action reads physical file bytes before and after; the collector records monotonic causal intervals, raw-byte hashes, a no-reset state lineage, and a signature. The evaluator's port-to-component mapping is stored apart from native input.

An independent verifier checks the Ed25519 signature, collector and executable hashes, state bytes, action effect, sensor projection, port inventory, and chronological lineage. A tampered capture is rejected. The native replay produces four initial *NO_PATH* decisions, a two-port *CRYSTAL* after the same-action control and comparator, *REVOKED* on the first c , four *NO_PATH* decisions during fresh-epoch accumulation, and a one-port *CRYSTAL* after the final comparator. A wrong checkpoint digest and forged epoch are rejected; full recovery succeeds in a fresh process.

At the fifth event, a and b have already been observed; the selector chooses c from the catalogue $\{a,b,c\}$. The sixth physical event actually fractures the class. This alignment between selection and outcome is an observed consequence of this protocol, not a promise that the selector always picks a separator. The withheld edit-transport panel contains 60 positive, 60 negative, and 60 wrapper/address-permutation cases across substitution, insertion, deletion, and four Unicode alphabets. Its 900 prefixes are checked by the independent oracle.

Results and inference boundary

The experiment establishes an intervention-owned observational relation, its revocation under actual counterevidence, and reconstruction from fresh evidence. A semantic label or designated target is not required by the operator. The physical setup still supplies the port inventory and action affordances. Two entities with indistinguishable responses to every supplied action remain one observational class until new experience separates them. This is a precise and useful referent mechanism, not a claim that the universe of possible separators was exhausted by the experiment.

The full V4 cost protocol measures the changed-atom transport against the preceding offset-based class rule at equal outcome. The median ratio at 64 events is 1.012 and peak-allocation ratio is 1.032, below preregistered 1.2 limits. These are local panel costs, not end-to-end production throughput. The final source package reproduces the relevant runtime and tests from extracted bytes and an isolated wheel.

Independent checking and attack surface

Four layers are checked separately. The Node verifier checks that the physical capture could have resulted from the recorded state transitions and that the signed body was not modified. The Python raw-contrast checker checks edit kind and all alignments. The class checker rebuilds every phase from original events, including epoch boundaries. The selector checker recomputes action coverage and deterministic choice. Agreement of all four is materially stronger than a test that compares only the final selected port with an evaluator label.

The negative controls illustrate different failure modes. The older signed panel with no same-action target control remains *NO_PATH*; it cannot be rescued by post-hoc threshold adjustment. A cross-format positive that failed the offset rule motivated a changed-atom rule, but a different changed atom at the same offset remains negative. Forged phase or epoch fields are rejected against the raw history. Duplicate available actions are rejected before selector ordering. A catalogue containing only already tried actions returns *NO_UNTRIED_ACTION*; adding an untried action changes the available experiment, not past evidence. These controls bind the precise scope of the positive physical result.

3.4. Deterministic resolution and continual revision

Falsification and result classes

The experiments separate logical completeness, actual feedback, and systems transport. In V116, a forged universal-input certificate is rejected; a past-only correlated context cannot erase the later separating experience; and unsupported guards remain acquisition obligations. Workspace and source-extracted audits each execute nine signed actual transitions across context exclusion, two-step context-changing goals, prepend, literal interleaving, distinguishing acquisition, counterevidence, and a newly acquired command. The decision records 18 transitions and cold recoveries across both audits, 10 observed goals, two quantified exclusions, four retained replans, two counterevidence interruptions, and two unique context discoveries. The 34 captures preserve native input under evaluator-label substitution. These are counts of a declared panel, not an extrapolated frequency in arbitrary worlds.

The V106 continuation API reports seven focused ownership/failure tests, 649 extracted regressions, 159 installed tests, and six physical transactions through actual cycles. V119 reports 40 focused tests, 957 source regressions, 467 installed tests, four actual transitions and commits, four cold recoveries, and eight label-invariant captures for its integration scope. The present source distribution runs its own complete phase and distribution checks from extracted bytes. These records should be read in their version order: later modules do not rewrite historical results.

Negative cases are essential. A no-path result after an unsupported guard is not counted as successful generalization. A changed action outcome is not silently normalized into the predicted state. A partial frontier produced by a resource ceiling is not rebranded as complete resolution. The paper's decision categories preserve these distinctions so that further learning can target a concrete missing witness.

Reproducibility and relation to NINMENI

NINMENI contributes the immutable NMU event substrate (Leunufna, 2026a) and a research discipline that separates organization, memory, material update, and output capability (Leunufna, 2026c; 2026d). TGI's guarded-law, frontier, actual-feedback, and atomic knowledge operators are its own derivations. The relevant modules are *tgi/guarded_word_closure.py*, its independent checker, *tgi/guarded_word_goal_frontier.py*, *tgi/guarded_word_goal_workflow.py*, and *tgi/local_acquisition_runtime.py*. Contracts and receipts identify the source scopes and exact commands. The publication bundle contains the article, source archive, frozen controls, and SHA-256 index.

Continual-learning protocol and failure preservation

Continual learning is a chain of transactions, not a single successful append. The V106 cycle API checks owned input/output, invalid plan, valid refusal, atomic failure, external digest, scope upgrade, and exact work boundaries. Six physical transactions include no-reset rollovers and an unexpected outcome. V116 then tests actual separating experience, counterevidence interruption, retained replans, and a newly learned command under signed capture. V119 re-bases local acquisition on the current native source while preserving the previous runtime modules. These are complementary rather than interchangeable tests: API recovery does not prove a context law, and an operator proof does not prove an atomic checkpoint.

The negative ledger is part of the method. A past-only prefix cannot resolve two continuations after the same prefix without another witness; false certainty there is a falsifier, not a minor quality reduction. Similarly, a completed test with its input instrument later found invalid must be retracted as an instrument failure. A resource exhaustion during proof construction yields no partial certified answer. Preservation of these cases is necessary to tell whether a later mechanism actually closes an obligation or merely hides it behind a new name.

3.5. Owned spatial execution and scale

Results

The terminal V11 measurement reports that all eight median append comparisons satisfy the frozen improvement criterion and all eight replay medians remain within the 20% guard. At 256 sources, append medians are 7.496 s for the spatial panel and 9.903 s for the temporal-phase panel, ratios 0.55 and 0.57 to their respective old baselines. Peak allocations reported for these two 256-source panels are 153,887,631 and 158,527,276 bytes. The reported values are process measurements for a declared CPU workload. They are not extrapolated to arbitrary source counts or a GPU implementation.

The local edit-transport operator used by the interventional relation analysis has a separate frozen cost criterion. Relative to the previous response-class implementation at a 64-event equal-outcome workload, its median time ratio is 1.012 and peak allocation ratio is 1.032, both below a declared 1.2 ceiling. This is not the V11 panel and cannot be averaged with it. It answers a narrower question: whether wrapper-invariant NMU edit transport adds excessive local overhead.

The full source distribution contains 247 runtime modules in the V71R paper binding. Its six gates check required material, hash parity, unexpected evidence exclusion, historical virtual-environment exclusion, raw phase from extraction, and distribution from extraction. Wheel tests for the physical fracture and selector run in the isolated installed environment. These checks establish that measurements and mechanisms are not dependent on an accidental workspace import path.

Bulk-native interpretation and negative controls

Bulk-native computation means reorganizing **execution** while retaining every character's static identity, source ownership, event order, and resolution target. It does not mean pooling characters into a new latent identity. V11 improves one cost center without changing the mechanism's mathematical meaning. The proof-bearing append remains more expensive than an unchecked in-memory update because it preserves source history and verifies a complete certificate. The correct economic comparison therefore has at least three independent axes: event throughput, proof/replay cost, and functional decision parity. A fast transition with an unverified partial certificate is a failed implementation, even if it returns a plausible sentence.

The retained failures are informative. Previous allocation repair work encountered host-memory *MemoryError* during an oversized verifier copy; localizing and removing the unnecessary copy allowed CPU recovery without reducing the proof. The V10 profile itself is not a speed result, and the V4 cost measurement does not prove V11 scale behavior. Preserving those separate experimental roles prevents selection of a convenient number as the whole-system result.

Full locked-panel results

The following table reproduces the median append and ratio fields from the terminal V11 protocol . Ratios compare each V11 point only with its matching frozen V10 baseline; they are not normalized across panels.

Panel	Sources	Median append (s)	V11/V10 append	V11/V10 replay	Checkpoint bytes
spatial	32	0.491	0.815	0.976	229,563
spatial	64	1.102	0.756	0.969	457,829
spatial	128	2.646	0.655	1.024	914,367
spatial	256	7.496	0.545	1.014	1,827,865
temporal phase	32	0.635	0.875	0.999	335,895
temporal phase	64	1.374	0.761	0.988	669,222
temporal phase	128	3.335	0.667	0.998	1,335,885
temporal phase	256	9.903	0.573	1.019	2,669,846

The trend is not represented by a single global speedup. Within this panel, the append ratio improves as source count rises, while replay ratios remain near one. That pattern is consistent with removal of repeated full-certificate copying during append, whereas complete replay still performs its full proof. The profiler diagnosis and this table together support the causal explanation for the matched implementation; they do not measure every possible input distribution. The checkpoint sizes increase with source count, which is expected because the original history and proof are preserved rather than compressed away.

Reproduction protocol and interpretation rules

A reproducer should verify the hash-bound baseline, use the same source counts and 256-NMU frame size, run the three untraced repetitions separately from the diagnostic traced run, then compute medians from the raw files. It should run functional parity and cold recovery before interpreting a timing improvement. Finally it should compare the source-extracted runtime hashes and installed wheel behavior with the workspace run. Changing the frame size, source composition, proof budget, or checker invocation creates a different experiment and cannot be silently pooled with this table.

The larger public evidence bundle is 1.49 GB compressed and 16.8 GiB in manifest-bound raw files. Its verifier reads every ZIP entry and compares its SHA-256 with the embedded index . A compact PDF alone therefore is not asked to carry all raw evidence; it identifies the immutable artifact needed to reproduce each number. This is the systems analogue of NINMENI's deposit-package practice: a paper explains the mechanism and result, while exact bytes and falsifiers remain inspectable.

4. Discussion

The five mechanism families address distinct obligations of the same TGI chain. Exact reconstruction tests source-lane isolation and route closure; phase controls test admissible structural organization; intervention tests whether a relation survives changes in observed physical state; resolution tests complete alternatives, feedback and recovery; and scale experiments test the cost of preserving these same decisions. A matching output without its source and certificate cannot substitute for any of these tests.

The experiments distinguish no path, unresolved alternatives and an incomplete search. Counterevidence is retained so a prior conclusion can be revised. A deterministic answer is therefore a decision under declared experience and scope, rather than a universal guarantee about unobserved world states. The signed physical traces test the specified intervention interface; hidden physical identity remains underdetermined when different referent assignments induce the same complete observations. This is an observability condition to test, not a label to inject into training.

5. Reproducibility, data and code availability

The public implementation and selected regression tests are available at <https://github.com/rfls-emyton/TGI/tree/de95c4c> and in the *tgi-foundation* version *0.2.0.dev0* package at <https://pypi.org/project/tgi-foundation/0.2.0.dev0/>. This code revision binds the methods and results reported here. Later public M2 releases have a separate version map and are not silently substituted for the paper’s tested code. The original evidence archive, including raw intervention captures, source receipts and locked scale measurements, is retained by the author. The public repository supplies source and selected tests; records that are not deposited there must be requested from the author for independent reanalysis. Local paths and contract filenames are provenance identifiers, not literature citations.

The experimental protocols require preservation of original Unicode IDs, event order, source scope, phase receipts, negative controls, exact version and independent certificate checking. Timing comparisons require the same locked input panel and output parity before a speed ratio is interpreted.

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